

SOLARGRID AI — Synthetic Engineering Corpus Index

SG-INDEX-000

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

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Document Title	SOLARGRID AI — Synthetic Engineering Corpus Index
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Provide document navigation, requirement-ID mapping, RAG keywords, cross-document relationships, and shared

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Scope: Provide document navigation, requirement-ID mapping, RAG keywords, cross-document relationships, and shared synthetic parameters.

Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

Table of Contents

- 1. 1. Corpus Purpose
- 2. 2. Document Register
- 3. 3. Requirement ID Map
- 4. 4. RAG Keyword Map
- 5. 5. Cross-Document Evidence Relationships
- 6. 6. Shared Synthetic System Parameters
- 7. 7. Replacement Strategy

1. Corpus Purpose

This index is the navigation layer for the SOLARGRID AI synthetic engineering corpus. It identifies the nine PDFs, their purposes, major topics, requirement-ID families, and cross-document relationships.

The corpus is synthetic and non-regulatory. It is designed for RAG retrieval, evidence linking, SafetyAgent demonstrations, PlannerAgent integration tests, monitoring/impact workflows, and Tool 16 testing.

2. Document Register

Document ID	File	Title	Primary Purpose
SG-OP-001	01_Synthetic_Grid_Operating_Code.pdf	Synthetic Grid Operating Code	Operating states and system-level limits
SG-GEN-002	02_Generator_Technical_Requirements.pdf	Generator Technical Requirements	G1/G2/G3 technical and commitment constraints
SG-BESS-003	03_BESS_Engineering_Manual.pdf	BESS Engineering Manual	Battery power, energy, SOC, reserve and safety
SG-FCST-004	04_Renewable_Forecast_Policy.pdf	Renewable Forecast and Forecast Error Policy	Solar forecast, uncertainty and replanning
SG-NET-005	05_Network_Power_Flow_Limits.pdf	Network Power Flow Engineering Limits	Voltage, line, transformer and contingency checks
SG-RES-006	06_Operating_Reserve_Procedure.pdf	Operating Reserve and Reliability Procedure	Reserve definitions, calculation and shortfall response
SG-MON-007	07_Plan_Change_Impact_Procedure.pdf	Plan Change Impact Assessment Procedure	Monitoring, thresholds and replan triggers
SG-SAFE-008	08_Engineering_Validation_Safety_Procedure.pdf	Engineering Validation and Safety Evidence Procedure	SafetyAgent validation and evidence handling

3. Requirement ID Map

ID Range / Family	Topic	Primary Document	Related Evidence
GRID-001...GRID-007	System operating states and overall constraints	SG-OP-001	RES, NET, BESS, GEN, SAFE
GEN-001...GEN-019	Generator limits, ramping, commitment and maintenance	SG-GEN-002	SG-OP-001, SG-RES-006, SG-MON-007
BESS-001...BESS-025	BESS energy, power, SOC, reserve, temperature and safety	SG-BESS-003	SG-OP-001, SG-RES-006, SG-SAFE-008
FCST-001...FCST-010	Solar forecasting and forecast error	SG-FCST-004	SG-RES-006, SG-MON-007
NET-001...NET-014	Voltage, thermal and power-flow limits	SG-NET-005	SG-OP-001, SG-RES-006, SG-SAFE-008
RES-001...RES-013	Operating reserve and shortfall logic	SG-RES-006	SG-OP-001, SG-GEN-002, SG-BESS-003
MON-001...MON-014	Monitoring and plan-change thresholds	SG-MON-007	SG-FCST-004, SG-RES-006, SG-SAFE-008
SAFE-001...SAFE-020	Validation, evidence, UNKNOWN and fail-safe behavior	SG-SAFE-008	All engineering documents
APP-001...APP-002	Human approval	SG-OP-001	SG-MON-007, SG-SAFE-008
VAL-001...VAL-002	Planning and validation evidence	SG-OP-001	SG-NET-005, SG-BESS-003, SG-SAFE-008

4. RAG Keyword Map

Keyword Cluster	Representative Terms
Battery	BESS, battery energy storage system, SOC, state of charge, charge limit, discharge limit, reserve SOC, battery reserve, energy
Generators	G1, G2, G3, Pmin, Pmax, ramp-up, ramp-down, minimum up-time, minimum down-time, startup, reserve, efficiency, fuel cost
Reserve	operating reserve, spinning reserve, non-spinning reserve, BESS reserve, generator reserve, reserve shortfall, headroom, commitment
Network	bus voltage, pu, line loading, transformer loading, thermal limit, contingency, power flow, overload, topology
Forecast	solar PV, renewable forecast, day-ahead, intra-day, forecast error, uncertainty, confidence interval, material change
Monitoring	baseline plan, system snapshot, material change, threshold crossing, plan validity, impact assessment, replan
Safety	SafetyAgent, engineering evidence, deterministic validation, PASS, FAIL, UNKNOWN, fail-safe, human review, approval

5. Cross-Document Evidence Relationships

Battery evidence: SG-BESS-003 provides physical BESS limits; SG-OP-001 uses those limits in operating requirements; SG-RES-006 uses them for reserve eligibility; SG-MON-007 defines the 10 MW BESS availability change trigger; SG-SAFE-008 defines how missing BESS evidence becomes UNKNOWN.

Generator evidence: SG-GEN-002 defines G1/G2/G3 limits and commitment constraints; SG-RES-006 calculates generator reserve; SG-FCST-004 explains why forecast changes may require additional generation; SG-MON-007 defines generator availability as a monitored change; SG-SAFE-008 validates generator feasibility.

Network evidence: SG-NET-005 defines voltage and thermal limits; SG-OP-001 summarizes system limits; SG-RES-006 warns that reserve satisfaction does not prove network feasibility; SG-MON-007 treats topology changes as mandatory impact assessments; SG-SAFE-008 requires current network evidence.

Safety evidence: SG-SAFE-008 is the principal SafetyAgent document, but it intentionally depends on the engineering evidence in SG-BESS-003, SG-GEN-002, SG-NET-005, SG-RES-006, SG-FCST-004, and SG-MON-007.

6. Shared Synthetic System Parameters

Parameter	Value
Peak synthetic load	235 MW
Minimum synthetic load	95 MW
Solar PV nameplate	180 MW
BESS usable energy capacity	120 MWh
BESS max continuous discharge	50 MW
BESS max continuous charge	40 MW
BESS normal SOC range	20–90%
BESS emergency reserve SOC	15% of usable capacity
Minimum operating reserve	25 MW
Normal bus voltage	0.95–1.05 pu
Alert bus voltage	0.93–1.07 pu
Line/transformer continuous limit	100% of synthetic rating
Planning thermal alert	90% of synthetic rating

7. Replacement Strategy

Each PDF has a stable document ID, explicit requirement IDs, configurable parameter tables, a synthetic status, and a clear non-regulatory footer. A real deploying organization can replace each document independently with an approved engineering source while retaining the RAG metadata structure.

INDEX-001 — Synthetic documents shall be treated as demonstration evidence only and shall not be used as operational authority.

INDEX-002 — Replacement with approved company documents should preserve traceability from document identity to individual engineering requirements where practical.